

## **CHRONIC KIDNEY DISEASE**

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## **INTRODUCTION**

Chronic kidney disease (CKD) is defined as abnormalities of kidney function or structure persisting for at least three months, with implications for health. This is a guide to optimise and to manage patients with chronic kidney disease to the point of referral and management by the nephrologist.

Haematuria and proteinuria are the hallmarks of glomerular disease. In addition, hypertension, impaired kidney function and fluid retention can be present to varying extents. The nature and severity of the underlying glomerular injury often dictate the nature and severity of these symptoms.

Conditions covered include (a) Chronic Glomerulonephritis (presenting as nephritic or nephrotic syndromes), (b) Nephropathies (e.g., secondary to underlying diabetes or other conditions) and (c) Chronic Kidney Diseases (with or without known underlying aetiology).

## **Importance**

Timely CKD detection and management play a major part in slowing down or preventing progression to kidney failure or other complications. Early detection is particularly significant given that patients are asymptomatic in the early stages of CKD. Primary healthcare professionals play an essential role in identifying patients at increased risk of CKD to detect it early.

## **Epidemiology**

In 2017, the estimated global prevalence of CKD was 9.1%. In Singapore, the prevalence of CKD among residents aged 18 to 74 years was 8.8% in 2019–2020, and CKD has remained in the top ten causes of death from 2009 to 2019 with CKD-related deaths rising by 76% within that decade.

## **Background**

1. The need to optimise siting of care for patients with chronic kidney disease (CKD) across the spectrum of healthcare services.
2. The burden of chronic kidney disease and end-stage renal failure (ESRF) is increasing over the years with eventual impact on the demand for renal replacement therapy services and overloading of the Nephrology Units in secondary and tertiary care.
3. Chronic kidney disease is a major public health problem. Adverse outcomes of chronic kidney disease can be prevented through early detection and treatment.

## **SCREENING AND DIAGNOSIS**

### **Screening of Asymptomatic Individuals**

Identify patients at increased risk of CKD by assessing their risk factors and determining the need to evaluate for CKD.

Risk Factors for CKD include:

- Age ( $\geq 60$  years)
- Gender (male > female)
- Diabetes mellitus
- Hypertension
- Obesity ( $\text{BMI} \geq 27.5 \text{ kg/m}^2$ )
- Hyperuricaemia or gout
- Smoking
- Family history of CKD or ESRF
- Hereditary kidney disease
- History of AKI
- Recurrent kidney stones
- Nephrotoxic medications (including frequent or chronic NSAID use)

### **Diagnosis**

Diagnosis of CKD is made if any of the following is present for at least three months:

- $\text{GFR} < 60 \text{ mL/min/1.73m}^2$
- $\text{UACR} \geq 3 \text{ mg/mmol}$  ( $\geq 30 \text{ mg/g}$ )
- Other marker of kidney damage

This can be done by adding “Chronic Renal Failure” to patient’s diagnoses and issue list.

## Stages of Chronic Kidney Disease (KDIGO CKD Definition)

| Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012        |     |                                  |       | Persistent albuminuria categories<br>Description and range |                             |                            |
|-----------------------------------------------------------------------|-----|----------------------------------|-------|------------------------------------------------------------|-----------------------------|----------------------------|
|                                                                       |     |                                  |       | A1                                                         | A2                          | A3                         |
|                                                                       |     |                                  |       | Normal to mildly increased                                 | Moderately increased        | Severely increased         |
|                                                                       |     |                                  |       | < 30 mg/g<br>< 3 mg/mmol                                   | 30–300 mg/g<br>3–30 mg/mmol | > 300 mg/g<br>> 30 mg/mmol |
| GFR categories (ml/min/1.73 m <sup>2</sup> )<br>Description and range | G1  | Normal or high                   | ≥ 90  |                                                            |                             |                            |
|                                                                       | G2  | Mildly decreased                 | 60–89 |                                                            |                             |                            |
|                                                                       | G3a | Mildly to moderately decreased   | 45–59 |                                                            |                             |                            |
|                                                                       | G3b | Moderately to severely decreased | 30–44 |                                                            |                             |                            |
|                                                                       | G4  | Severely decreased               | 15–29 |                                                            |                             |                            |
|                                                                       | G5  | Kidney failure                   | < 15  |                                                            |                             |                            |

Green, low risk (if no other markers of kidney disease, no CKD); yellow, moderately increased risk; orange, high risk; red, very high risk.  
GFR, glomerular filtration rate.

Source: *Kidney Disease Improving Global Outcomes (KDIGO). Clinical Practice Guideline for Acute Kidney Injury. March 2012*

The calculation for GFR using EPI is as follows:

$$\text{GFR} = 141 \times \min(\text{Se Creat}/\kappa \text{ or } 1)^\alpha \times \max(\text{Scr}/\kappa \text{ or } 1)^{-1.209} \times 0.993^{\text{Age}} \times 1.018 [\text{if female}] \times 1.159 [\text{if black}]$$

Where: *Se Creat* is serum creatinine in mg/dL,  
 $\kappa$  is 0.7 for females and 0.9 for males,  
 $\alpha$  is -0.329 for females and -0.411 for males,  
*min* indicates the minimum of *Se Creat*/ $\kappa$  or 1, and  
*max* indicates the maximum of *Se Creat*/ $\kappa$  or 1.

Kidney damage is defined as pathologic abnormalities or markers of damage, including abnormalities in blood or urine tests or imaging studies.

### **Principles of ACR / PCR**

ACR - more sensitive for Microalbuminuria, to detect & identify proteinuria.

PCR - more sensitive for Macroalbuminuria, to monitor & quantify proteinuria (for Non-DM pts & also for DM pts with ACR > 30)

### **HALT-CKD Criteria**

Normal ACR for male is < 2.5 mg/mmol, normal ACR for female is < 3.5 mg/mmol

Male Microalbuminuria: ACR is 2.5 to 30 mg/mmol, Female Microalbuminuria: ACR is 3.5 to 30 mg/mmol

Male & Female Macroalbuminuria: ACR is > 30 to 70 mg/mmol

Male & Female Overt Proteinuria: ACR is > 70 mg/mmol

ACR 30 mg/mmol is equivalent to PCR 50 mg/mmol & this is equivalent to UTP 500 mg/day

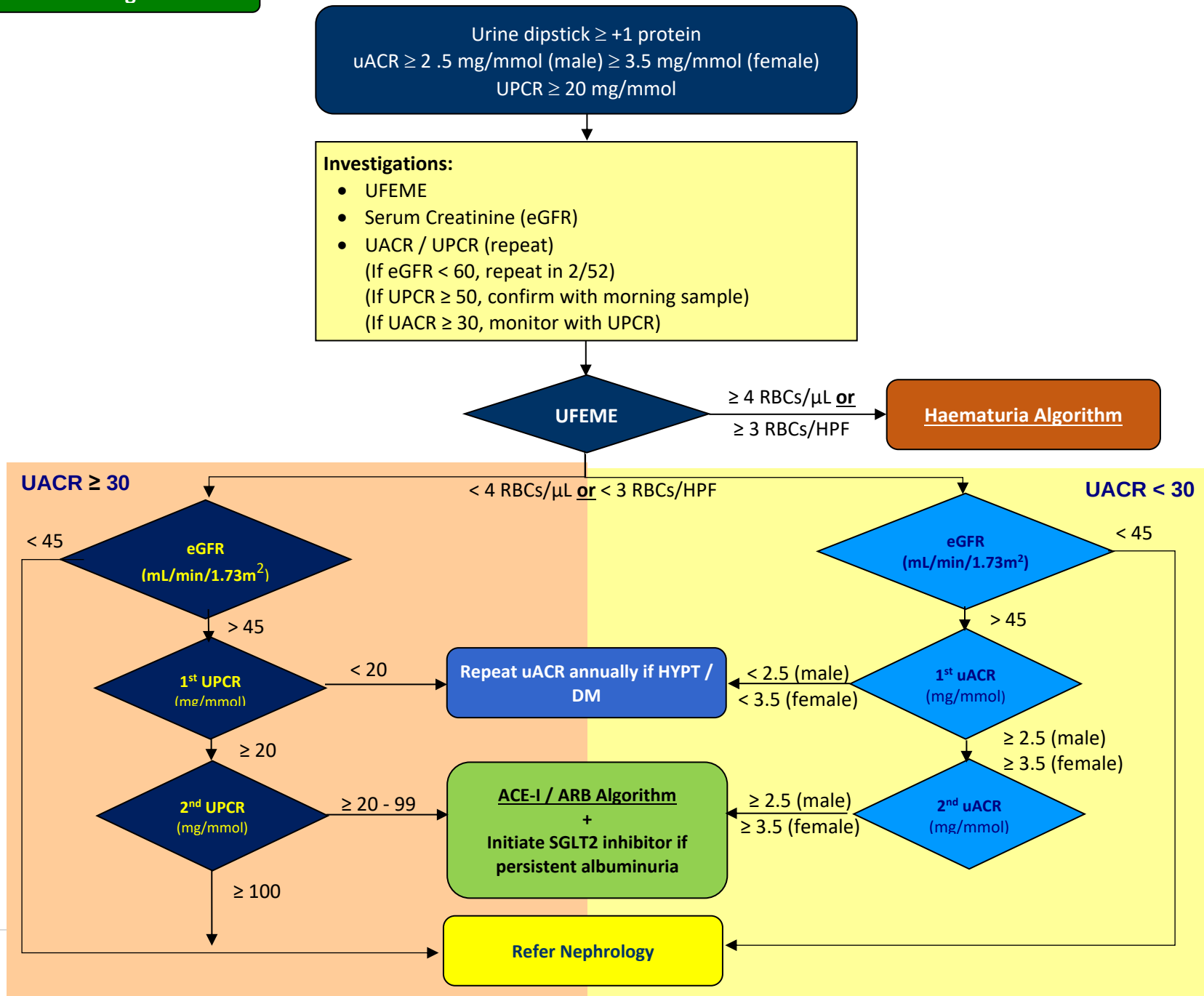
ACR 70 mg/mmol is equivalent to PCR 100 mg/mmol & this is equivalent to UTP 1,000 mg/day or 1 g/day

The normal PCR for 16 years old & above is < 20 mg/mmol. We can use PCR to replace 24 hr UTP. Just need to multiply PCR by 10 to get the 24 hr UTP in mg/day.

### **Common Causes of CKD & ESRF**

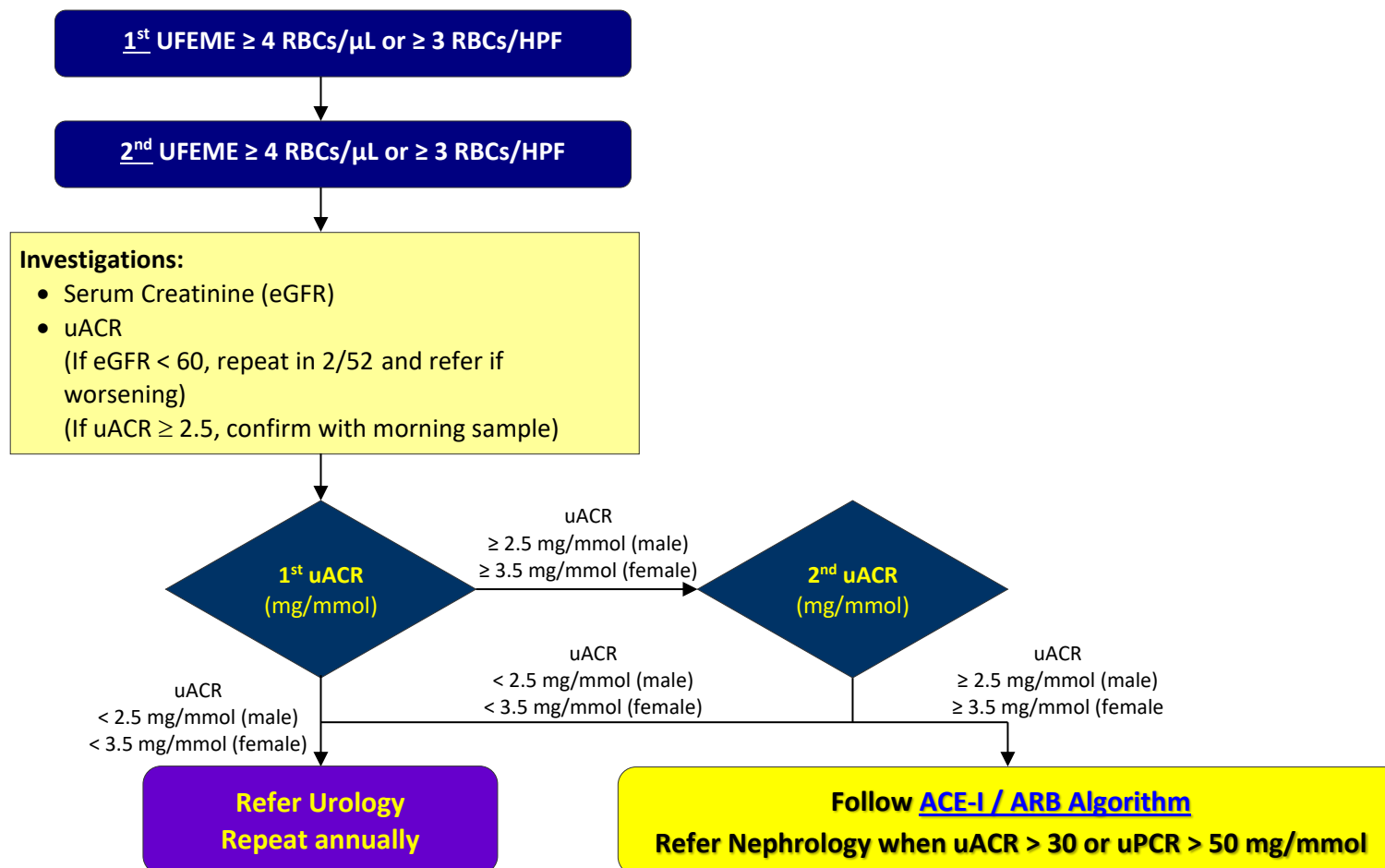
1. Diabetic kidney disease
2. Hypertensive nephrosclerosis
3. Primary glomerulonephritis (GN)
4. Autoimmune diseases - SLE
5. Cystic diseases - polycystic kidney disease
6. Others - chronic pyelonephritis, obstruction

## Proteinuria Algorithm



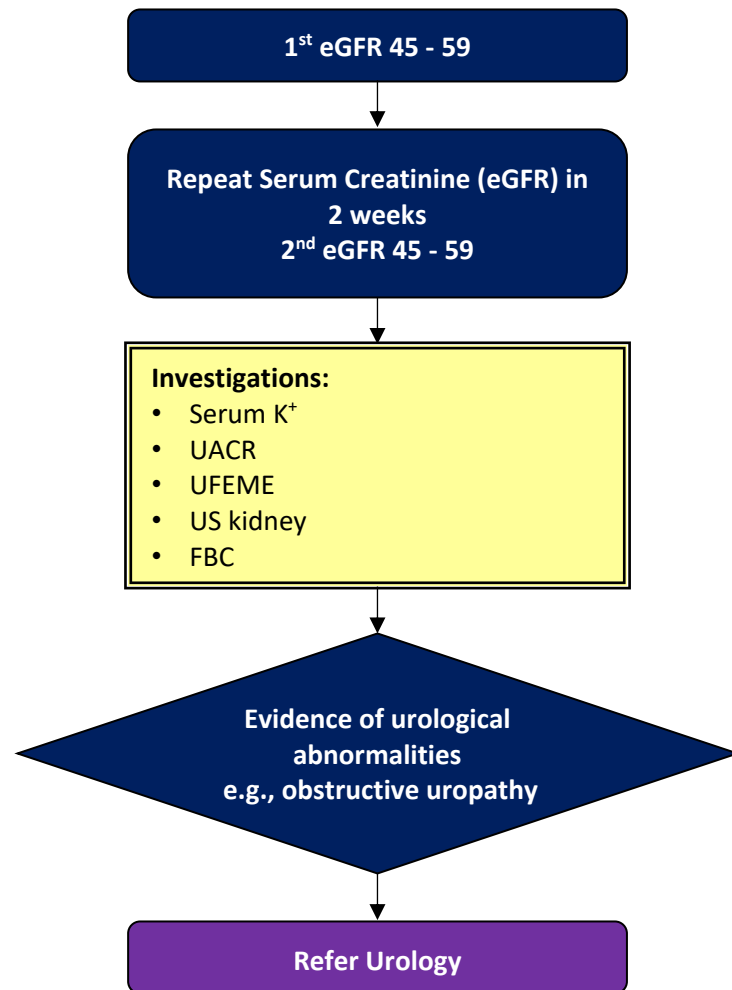
## Haematuria Algorithm

(Refer to Haematuria care path as well)



\* Please note that if a urine dipstix shows blood 1+ or more, UFEME needs to be ordered for further evaluation. If UFEME shows at least 4 RBCs/μL, Haematuria Algorithm applies.  
After Urological clearance, only significant increase in number of RBCs/μL needs a repeat review by Urology.

### CKD G3 Algorithm



#### Risk factor modifications:

- Start and optimise ACE-I/ARB
- Start SGLT2 inhibitor if persistent albuminuria, can be started even when dose of ACEI/ARB is not maximized

#### CKD Complications:

##### Anaemia

- Check Hb, ferritin annually
- If Hb < 10 g/dL, ferritin < 100 µg/L, work up for causes of iron deficiency anaemia (e.g., stool occult blood), consider iron repletion
- If Hb < 10 g/dL, ferritin ≥ 200 µg/L, refer Nephrology

Refer Nephrology

#### Monitoring Progression

- Refer if eGFR declines > 10 ml/min/1.73m<sup>2</sup> per year or > 5ml/min/1.73m<sup>2</sup> over 3 months
- Refer if eGFR < 45 ml/min/1.73m<sup>2</sup>

## MANAGEMENT

### Targets of Treatment (HALT CKD)

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Diagnose</u> CKD</b>                             | <ul style="list-style-type: none"> <li>Add “Chronic Renal Failure” as visit diagnosis and to problem list if UACR <math>\geq 3</math> mg/mmol or eGFR <math>&lt;60</math> mL/min/1.73m<sup>2</sup> for more than 3 months apart</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                |
| <b><u>Lifestyle</u><br/><u>Modification</u></b>        | <ul style="list-style-type: none"> <li>Refer all patients age <math>&lt; 80</math> years old for HALT-CKD counselling</li> <li>Stop smoking, refer smoking cessation clinic by pharmacist</li> <li>Encourage weight loss, refer dietician if BMI <math>&gt; 23</math> kg/m<sup>2</sup>, weight management clinic if BMI <math>&gt; 27.5</math> kg/m<sup>2</sup></li> <li>Counsel on low salt (<math>&lt; 2</math> g/day) diet</li> <li>Counsel on low protein diet (<math>&lt; 0.8</math> g/kg/day) and dietician review for CKD G3B patients without DM</li> <li>Advise on <b>150</b> min/week of moderate intensity exercise</li> </ul> |
| <b><u>Maximize</u><br/><u>ACE-I/ARB</u></b>            | Optimize dosages until one of these endpoints below: <ol style="list-style-type: none"> <li><b>Maximal recommended or tolerated dose</b></li> <li><b>Normoalbuminuria + Achieve BP target</b></li> <li>When starting or increasing ACE-I / ARB, order ACE-I / ARB panel in 2 – 4 weeks with CM review</li> </ol>                                                                                                                                                                                                                                                                                                                          |
| <b><u>Optimize</u> <u>BP</u></b>                       | <ol style="list-style-type: none"> <li><b><math>&lt; 130/80</math> mmHg</b> for ALL patients</li> <li><b><math>&lt; 140/90</math> mmHg</b> for older patients, high fall risk, multiple co-morbidities</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b><u>Optimize</u><br/><u>HbA1c</u></b>                | <ol style="list-style-type: none"> <li><math>\leq 7\%</math>: Age <math>\leq 75</math> years old</li> <li><math>\leq 8\%</math>: Age 76 – 80 years old</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b><u>Optimize</u> <u>LDL-C</u></b>                    | <ol style="list-style-type: none"> <li><b><math>&lt; 1.8</math> mmol/L</b> for DM patients</li> <li><b><math>&lt; 2.6</math> mmol/L</b> for non-DM patients</li> <li>More stringent targets for patients with ASCVD or additional risk factors</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                 |
| <b><u>Start</u> <u>SGLT-2</u><br/><u>Inhibitor</u></b> | <ol style="list-style-type: none"> <li>Can be started if patient is on an ACE-I/ARB; check Renal Panel in elderly, advanced CKD or on diuretic</li> <li>Multiple benefits such as weight loss, BP and DM control, reducing albuminuria, retarding progression, reducing mortality</li> </ol>                                                                                                                                                                                                                                                                                                                                              |
| <b><u>Co-manage</u><br/><u>with Renal</u></b>          | Refer <b>CKD G3B, G4 and G5</b> or persistent significant albuminuria to Nephrology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## Follow-up and Assessment of CKD Patients

|                                     | G1 or G2 with moderately increased albuminuria (A2) OR G3a with normal to mildly increased albuminuria (A1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | G3a with moderately increased albuminuria (A2) OR G1-2 with severely increased albuminuria (A3) | G3a with severely increased albuminuria (A3) |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------|
| Frequency of review                 | Consider following up every 6–12 months                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Consider following up every 3–6 months                                                          | Consider following up every 1–4 months       |
| Clinical and laboratory assessments | Selection of parameters for review should be tailored to the patients' CKD stage and clinical needs, such as comorbidities management. Clinical and laboratory assessments include: <ul style="list-style-type: none"><li>• Overall CV risk</li><li>• Blood pressure</li><li>• BMI</li><li>• Oedema (for patients with A3)</li><li>• eGFR and UACR (if UACR is &gt;30 mg/mmol, consider UPCR especially if non-albumin proteinuria is suspected)</li><li>• Biochemical profile including electrolytes, creatinine, and HbA1c for patients with DM</li><li>• Lipid profile</li><li>• FBC (for patients with CKD stage G3aA2 or G1-3A3)</li><li>• Albumin, calcium, and phosphate (for patients with CKD stage G3aA3)</li></ul> |                                                                                                 |                                              |

A, albuminuria category; BMI, body mass index, CV, cardiovascular; DM, diabetes mellitus; eGFR, estimated glomerular filtration rate; FBC, full blood count; G, GFR category; HbA1c, haemoglobin A1c; UACR, urine albumin: creatinine ratio; UPCR, urine protein: creatinine ratio

Source: ACE Clinical Guidance. Chronic Kidney Disease Management – Delaying progression and reducing cardiovascular complications. October 2023

## ACE-I / ARB Algorithm

UACR  $\geq 2.5$  mg/mmol (Male);  $\geq 3.5$  mg/mmol (Female)  
(at least 2 separate readings > 3 months apart for CKD G1 and G2)

SBP  $\geq 110$  mmHg

No

Yes

Optimise / Start ACE-I / ARB

Check serum K<sup>+</sup> / Cr in 2 - 4 weeks

$\uparrow$  Cr > 30%  
K<sup>+</sup> > 5.0 mmol/L  
CKD 3B with eGFR  $\downarrow$  > 4 mL/min/1.73m<sup>2</sup>  
BP < 110/70 mmHg

### Optimise to one of the targets below:

1. Achieved Normoalbuminuria at target BP, or
2. Maximal recommended dose of ACE-I / ARB, or
3. Maximal tolerated dose

SGLT2 inhibitor can be used as an adjunct to control albuminuria at any dose of ACE-I/ARB

### Referral to Nephrology:

1. DM: UACR > 70 mg/mmol or UPCR > 100 mg/mmol despite maximum dose
2. Non-DM: UACR > 70 mg/mmol or UPCR > 100 mg/mmol
3. CKD G3B and above

### SBP < 110 mmHg

1. Exclude acute causes
2. Stop other anti-hypertensive agents (in this order: CCB,  $\alpha$  blockers,  $\beta$  blockers, diuretics)

### Creatinine increase > 30%

1. Exclude dehydration, sepsis, uncontrolled BP, nephrotoxic drugs
2. Reduce ACE-I/ARB to previous dose
3. Consider RAS if Cr continue to rise > 30% despite above
4. If eGFR decline  $\geq 25\%$  from baseline – stop ACE-I/ARB, refer Nephrology

### K<sup>+</sup> > 5.0 mmol/L

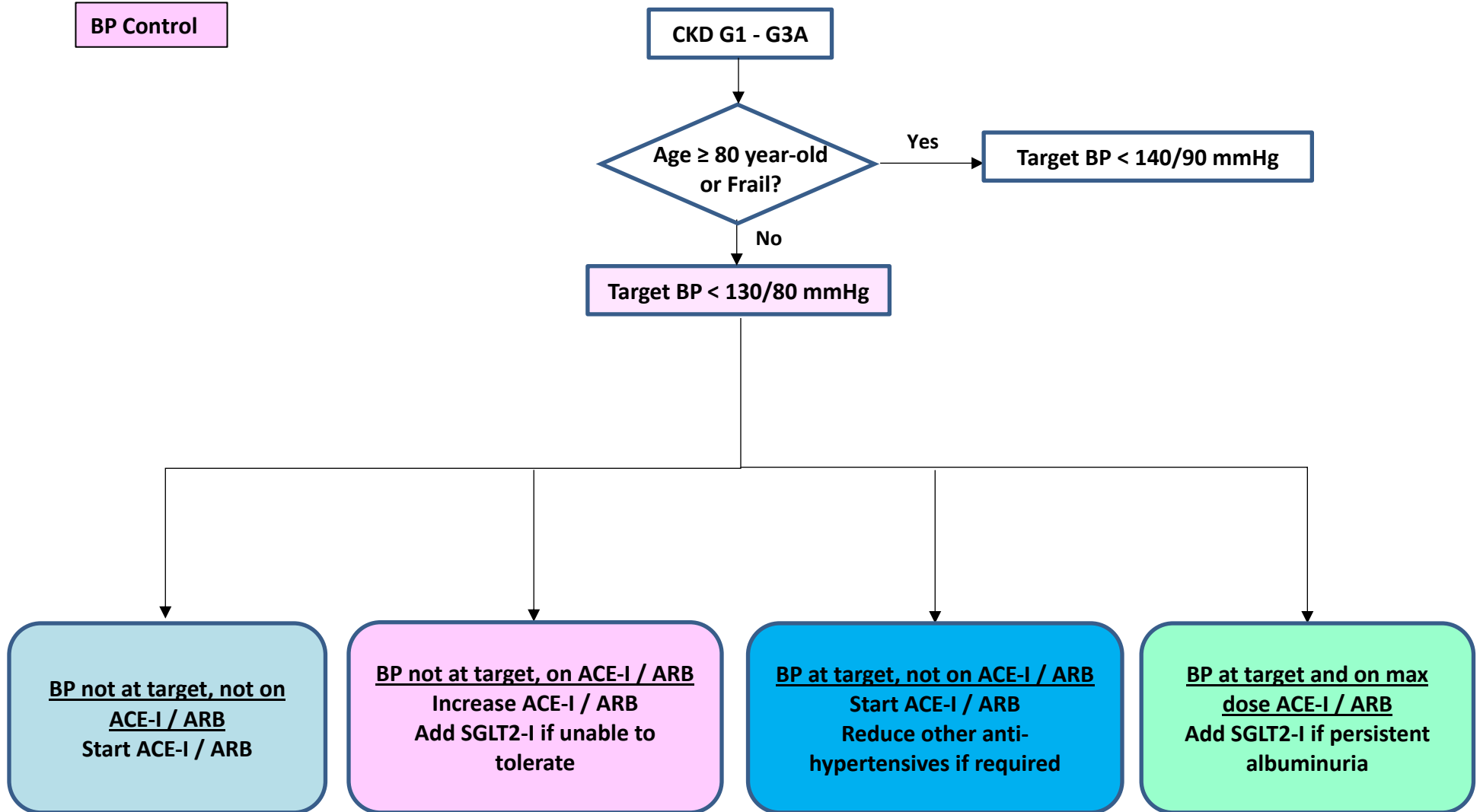
1. Advise on low K<sup>+</sup> diet
2. Add Thiazide diuretic if SBP  $\geq 120/70$  mmHg
3. If K<sup>+</sup> > 5.5 mmol/L despite above, reduce to previous dose and consider Resonium
4. If K<sup>+</sup>  $\geq 6.0$  mmol/L despite above, advise patient to attend ED for repeat serum K<sup>+</sup> and treatment

### CKD G3B onwards with eGFR decline > 5 mL/min/1.73m<sup>2</sup>

1. Dose adjust medications, refer Nephrology (early)
2. Recheck BP and increase other anti-hypertensives if BP above target (Diuretics if edema/ heart failure; CCB otherwise)

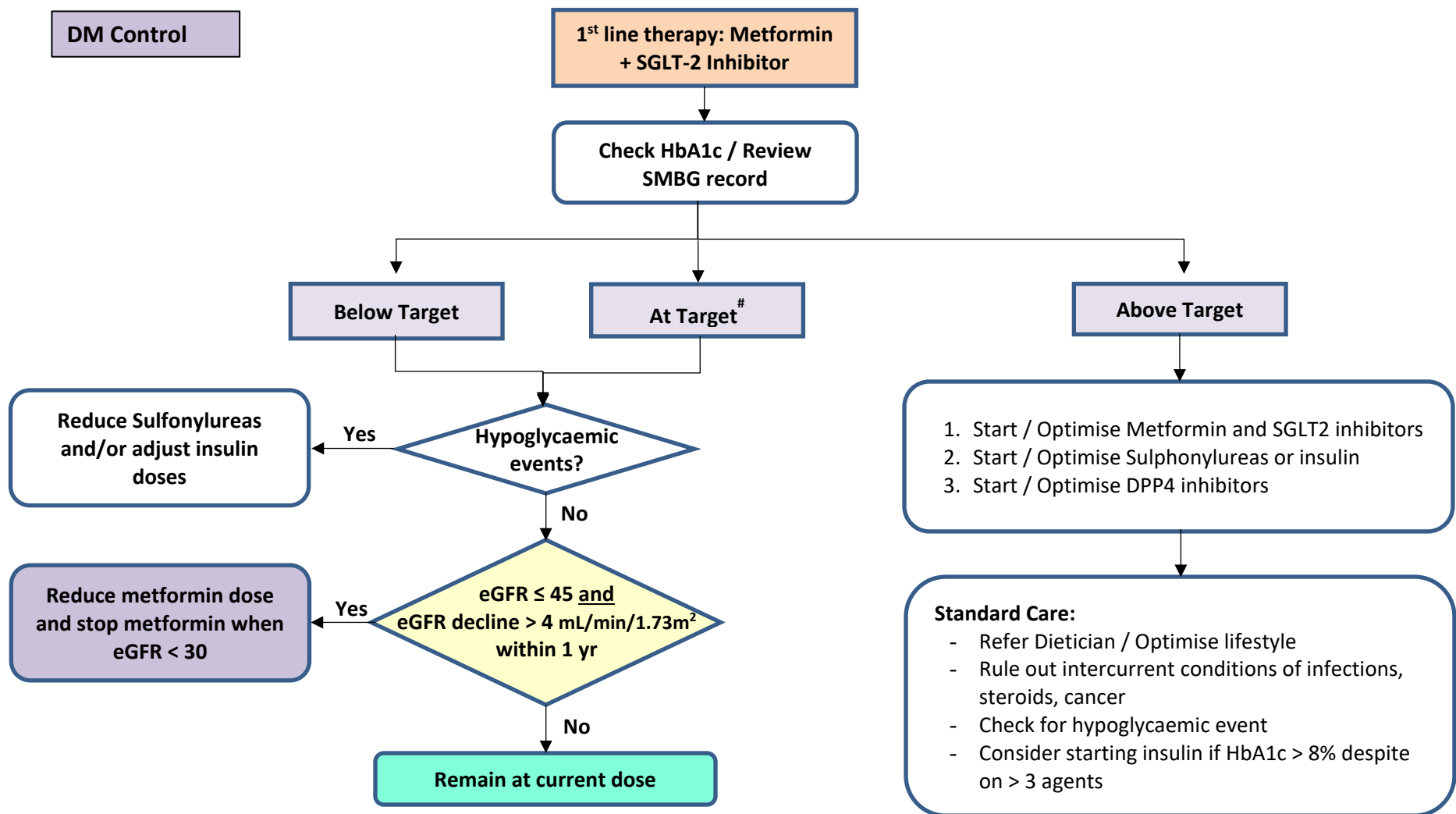
Proceed to BP Algorithm if BP target not achieved despite above

**BP Control**



Referral to Nephrology or General Medicine as indicated:  
BP > 150/90 mmHg despite 3 anti-hypertensives at maximal dose

## DM Control



### # Target HbA1c:

- 7% in most adults
- More stringent if short duration of DM, longer life expectancy, no significant CVD
- Less stringent if recurrent / severe hypoglycaemia, advanced micro / macro complications, advanced, advanced CKD (Stage 4), elderly ≥ 75 years-old

Reduce or stop long-acting sulphonylureas.  
Tail down Metformin when eGFR < 50 mL/min/1.73m<sup>2</sup> and stop when < 30 mL/min/1.73m<sup>2</sup>.

## Use of SGLT2 Inhibitors in CKD

SGLT2 inhibitors have been shown to reduce risk of worsening kidney function, onset of kidney failure or death from renal causes, with the added benefit of reducing risk of CV events in patients with CKD, with or without DM.

An acute eGFR decline may occur at 2–4 weeks after initiation of an SGLT2 inhibitor. In the absence of haemodynamic instability or an alternate cause of AKI, the initial rise in serum creatinine of up to 30% is not associated with long-term kidney function loss, and treatment with SGLT2 inhibitors should not be discontinued. Renal panel should be checked if concerns regarding AKI upon SGLT2 inhibitor initiation, such as concurrent diuretic therapy, advanced CKD, or in elderly patients.

In patients with CKD who do not have DM, the recommended dosage of Dapagliflozin and Empagliflozin is limited to 10 mg daily. This guidance is based on current evidence regarding the optimal balance of efficacy and safety in non-diabetic CKD patients.

Please ensure the following criteria are met:

1. Initiated on an ACE-I/ARB and appropriate eGFR to initiate
2. If significant proteinuria (TUP > 1 g/day) is present for patients without DM, to consider other etiologies and consider referral to Nephrology
3. Ensure patient has been adequately counselled (use SmartPhrase **.NUPDRSGLT2i**): Advised on action, benefits, need for adequate hydration, genital hygiene, sick day precaution, increased risk of urinary frequency, urinary and genital infection, euglycemic DKA and AOCKD.

## Starting dose of SGLT2 inhibitors

| Table 2. eGFR levels and starting dose for SGLT2 inhibitors for patients with CKD |                                                                   |                                          |                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                   | Canagliflozin                                                     | Dapagliflozin                            | Empagliflozin                                                                                                                                                                                                                  |
| eGFR level for initiation                                                         | eGFR $\geq 30$ mL/min/1.73m <sup>2</sup> or CrCl $\geq 30$ mL/min | eGFR $\geq 25$ mL/min/1.73m <sup>2</sup> | eGFR $\geq 20$ mL/min/1.73m <sup>2</sup> was used in the recent EMPA-KIDNEY trial with initial dose of 10 mg OD and included participants with or without T2DM. <sup>16</sup> Refer to product insert for updated information. |
| Starting dose                                                                     | 100 mg OD                                                         | 5–10 mg OD                               |                                                                                                                                                                                                                                |
| Registered indication                                                             | CKD and T2DM                                                      | CKD (+/- T2DM)                           |                                                                                                                                                                                                                                |

CKD, chronic kidney disease; CrCl, creatinine clearance; eGFR, estimated glomerular filtrationrate; EMPA-KIDNEY, The Study of Heart and Kidney Protection With Empagliflozin; OD, once daily; T2DM, type 2 diabetes mellitus

Source: ACE Clinical Guidance. Chronic Kidney Disease Management – Delaying progression and reducing cardiovascular complications. October 2023

## Angiotensin-Converting Enzyme Inhibitors (ACE-Is) & Angiotensin II Receptor Blockers (ARBs) dosage recommendations

| Medication Class | Medication Name      | Dose                                                                     |                                                                 |
|------------------|----------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------|
|                  |                      | Normal Renal Function                                                    | Renal Dose Adjustment                                           |
| ACE inhibitors   | Captopril            | Initial: 6.25-25mg BD-TDS<br>Max: 50mg TDS                               | CrCl 10-50: administer 75% of normal dose every 12-18 hours     |
|                  | Enalapril            | Initial: 2.5-10mg OD<br>Max: 40mg/day                                    | CrCl 10-30: Initial 2.5mg/day in 1-2 divided dose, Max 20mg/day |
|                  | Imidapril            | Initial: 2.5-5mg OD<br>Max: 20mg OD                                      | Initial 2.5mg OD (in patients with renal impairment)            |
|                  | Lisinopril           | Initial: 2.5-10mg OD<br>Max: 40mg OD                                     | CrCl 10-30: Initial 2.5-5mg OD                                  |
|                  | Perindopril arginine | Initial: 2.5-5mg OM<br>Max: 10mg OD                                      | CrCl 60-89: Max 5mg/day<br>CrCl 30-59: Max 2.5mg/day            |
|                  | Perindopril erbumine | Initial: 4mg OD<br>Max: 8-16mg OD                                        | CrCl ≥30-80: Initial 2mg OM, Max 8mg OM                         |
|                  | Ramipril             | Initial: 2.5mg OD<br>Max: 10mg/day                                       | CrCl 20-50: Initial 1.25 mg/day, Max 5 mg/day                   |
| ARBs             | Candesartan          | Initial: 4-8mg OD<br>Max: 32mg OD                                        | Initial 4mg OM (in patients with renal impairment)              |
|                  | Eprosartan           | Initial: 300-600mg OM<br>Max: 1200mg OD (in clinical trials for 8 weeks) | CrCl <60: Initial 300mg OM                                      |
|                  | Fimasartan           | Initial: 60mg OD<br>Max: 120mg OD                                        | CrCl 30-80: no initial dose adjustment needed                   |
|                  | Irbesartan           | Initial: 75-150 mg OD<br>Max: 300 mg OD                                  | No dose adjustment needed                                       |
|                  | Losartan             | Initial: 25-50mg OD<br>Max: 100mg OD                                     | No dose adjustment needed                                       |
|                  | Olmesartan           | Initial: 20mg OD<br>Max: 40mg OD                                         | No dose adjustment needed                                       |
|                  | Telmisartan          | Initial: 20-40mg OD<br>Max: 80mg OD                                      | No dose adjustment needed                                       |
|                  | Valsartan            | Initial: 80-160mg OD<br>Max: 320mg OD                                    | CrCl >10: no dose adjustment needed                             |

Medications in **bold** denote availability on government subsidy list. Information from local product inserts has been referenced, and supplemented with information from consolidated product monographs (e.g. Lexicomp) where local product inserts are unavailable or unclear. Refer to product inserts for full details before prescribing. Renal dose adjustment for CrCl <30 are excluded from the table.

Information from other references (e.g. international guidelines) may differ. Clinical judgement should be exercised at all times when making decisions for an individual patient.

ACE inhibitor, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blockers; CrCl, creatinine clearance in mL/min; BD, twice daily; OD, once daily; OM, once every morning; TDS, three times daily

Source: ACE Clinical Guidance. Chronic Kidney Disease Management – Delaying progression and reducing cardiovascular complications. October 2023

## Common medications that should be prescribed with caution in patients with CKD

| Antimicrobial                                                                                                                                                                                                                     | Cardiovascular                                                                                                                                                                                                                     | DM                                                                                                                                                                                                                                                                | Pain management                                                                                                                                    | Others                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Ceftazidime</li> <li>• Ciprofloxacin</li> <li>• Fluconazole</li> <li>• Nitrofurantoin</li> <li>• Sulfamethoxazole/trimethoprim</li> <li>• Terbinafine</li> <li>• Valaciclovir</li> </ul> | <ul style="list-style-type: none"> <li>• Apixaban</li> <li>• Atenolol</li> <li>• Dabigatran</li> <li>• Digoxin</li> <li>• Edoxaban</li> <li>• Fenofibrate</li> <li>• Rivaroxaban</li> <li>• Sotalol</li> <li>• Warfarin</li> </ul> | <ul style="list-style-type: none"> <li>• <i>DPP4 inhibitors except Linagliptin</i></li> <li>• <i>Insulin</i></li> <li>• Metformin</li> <li>• <i>PPAR<math>\gamma</math> agonist</i></li> <li>• <i>SGLT2 inhibitors</i></li> <li>• <i>Sulfonylureas</i></li> </ul> | <ul style="list-style-type: none"> <li>• Gabapentin</li> <li>• <i>NSAIDs</i></li> <li>• <i>Opioids analgesics</i></li> <li>• Pregabalin</li> </ul> | <ul style="list-style-type: none"> <li>• Colchicine</li> <li>• <i>First-generation antihistamines</i></li> <li>• Lithium</li> </ul> |

### Medications that should be ceased during illness and may resume when patients have recovered from sickness

#### SADMANS

- *Sulfonylurea, other secretagogues*
- *ACE inhibitors*
- *Diuretics, direct renin inhibitors*
- Metformin
- ARBs
- *Nonsteroidal anti-inflammatory drugs (NSAIDs)<sup>‡</sup>*
- *SGLT2 inhibitors or "flozins"*

Medication classes have been stated in *italics*.

ACE inhibitors, angiotensin-converting enzyme inhibitors; ARBs, angiotensin II receptor blockers; DM, diabetes mellitus; DPP4, dipeptidyl peptidase-4; eGFR, estimated glomerular filtration rate in mL/min/1.73m<sup>2</sup>; NSAID, non-steroidal anti-inflammatory drug; PPAR $\gamma$ , peroxisome proliferator-activated receptor gamma; SGLT2 inhibitors, sodium-glucose co-transporter 2 inhibitors

<sup>†</sup>List is not exhaustive; refer to package inserts for full details of medications.

<sup>‡</sup>Avoid over-the-counter use of medications containing NSAIDs.

Source: ACE Clinical Guidance. Chronic Kidney Disease Management – Delaying progression and reducing cardiovascular complications. October 2023

- Allopurinol may need to be dose-adjusted (refer to [Gout CPG](#))

## Other Management

### 1. Stop use of nephrotoxic drugs

- a) NSAID (except Aspirin)
- b) Antibiotics
  - i. Sulphonamides
  - ii. Aminoglycosides
- c) Contrast media

### 2. Diet advice for early CKD



#### Protein intake

Protein-restricted diets may not reduce kidney failure events nor increase GFR among patients with CKD G3b and earlier (although protein restriction may improve these outcomes in later stages of CKD).<sup>38</sup>



#### Potassium intake

Current evidence regarding the effect of potassium restriction on CKD progression is mixed,<sup>37, 38</sup> and hence potassium restriction is not recommended as a routine dietary intervention for all patients with CKD.

However, for patients on ACE inhibitor/ARB with hyperkalaemia, reducing potassium intake through diet may be required and it is usually preferred over stopping pharmacotherapy.



#### Sodium intake

Unless contraindicated, sodium restriction can be considered in order to reduce blood pressure<sup>39</sup>, albuminuria,<sup>39</sup> and risk of CV events.<sup>40</sup>

| Implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Contraindications                                                                                                                                                                                                                            | Precautions                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>A specific target intake for patients with CKD is not yet apparent from current evidence. However, the recommended intake of less than 2g of sodium per day (less than 1 teaspoon of salt) for the general population<sup>41</sup> has been adopted for patients with CKD in recent guidelines.<sup>23, 42</sup></li><li><b>Drastic reductions in sodium intake (e.g. to 1.4g sodium per day) is not advised</b> as it may result in reduced GFR, orthostatic hypotension, and hyperuricemia.<sup>43</sup></li><li>The sodium content of local foods can be checked using HPB's online tool (<a href="https://focos.hpb.gov.sg/eservices/ENCF/">https://focos.hpb.gov.sg/eservices/ENCF/</a>), noting that sodium content could vary between food vendors due to differences in preparation.</li><li>Due to the risk of hyperkalaemia, <b>the DASH diet and potassium salt substitutes may be inappropriate</b> for patients with advanced CKD,<sup>23, 42</sup> hyporeninemic hypoaldosteronism, or impaired potassium excretion.<sup>23</sup></li></ul> | <p>Sodium restriction is contraindicated for patients with CKD who:</p> <ul style="list-style-type: none"><li>Have salt-wasting nephropathy<sup>6, 23</sup> or</li><li>Are hypotensive or prone to volume contraction.<sup>6</sup></li></ul> | <ul style="list-style-type: none"><li>Monitor for adverse effects of sodium restriction, which may include symptomatic hypotension<sup>39</sup></li><li>Patients with orthostatic hypotension may require more individualised guidance on sodium intake.<sup>42</sup></li></ul> |

Source: ACE Clinical Guidance. Chronic Kidney Disease Management – Delaying progression and reducing cardiovascular complications. October 2023

## **Role of Health Team Members**

### **1. Family Physician**

- Able to provide all aspects of primary medical care from screening, diagnosis and management of CKD, including health promotion and prevention / treatment of complications
- Collaborates with other healthcare providers in the health team to provide holistic and patient-centred care

### **2. HALT-CKD Coordinator**

- Patient education on CKD and its management, including diet and lifestyle advice on sodium reduction, weight management and smoking cessation
- Refer patients to other services where applicable (dietitian, weight management programme and/or smoking cessation clinic)

### **3. Care Manager**

- Patient education on diabetes, HbA1c and glucose targets and lifestyle measures
- Teach home glucose and home blood pressure monitoring
- Patients who are stable and have achieved good glycaemic, BP or lipid level control may have alternate clinic visits between the doctor and the care manager.

### **4. Advanced Practice Nurse**

- Manage patients with diabetes, hypertension and dyslipidaemia within scope of practice
- Initiate and titrate medications in collaboration with physicians
- Manage common acute conditions such as upper respiratory tract infection, gastroenteritis should the patient concomitantly present with these reasons for encounter

### **5. Pharmacist**

- Smoking cessation clinic
- Detect, prevent and address drug-drug/drug-disease interactions
- Perform medication reconciliation
- **Recommend renal adjusted doses for medications**

## **6. Dietitian**

- Dietary management for patients with CKD
- Medical nutrition therapy for patients with co-morbidities such as DM / hyperlipidaemia / malnutrition with CKD
- Patient education on weight management (underweight or overweight) i.e. referral to Weight Management Programme, which includes Dietetic and Nutrition, and Physical Exercises to reduce weight over time

## **7. Medical Social Worker or Psychologist**

- Conduct biopsychosocial assessment
- Provide intervention for social assistance including application of schemes and services, and referral for community resources and caregiver support

## **8. Financial Counsellor**

- Financial counselling for patients with issues on payment of medical bills
- Advised on MediSave utilisation for chronic disease medical bills
- Application on government endowment fund - MediFund and Medication Assistance Fund for subsidies on high cost drugs
- Application on Institutional Medical Fund-Remission for eligible non-residents

## **9. Psychologist**

- Emotional and psychological support to manage psychological distress in patients with CKD
- Psychological and behavioural interventions to develop coping strategies to cope with the development and progression of kidney disease, improve disease management and quality of life
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders *etc.*)
- Psychological support for carers experiencing caregivers stress

## **10. Physiotherapist**

- Physical assessment to ascertain fitness
- Advise on suitable physical activities to reduce weight

## **SPECIAL SITUATIONS**

### **Acute Kidney Injury**

While evaluating patients with eGFR drop, be aware of the possibility of Acute Kidney Injury (AKI), defined as any of the following:

- Increase in serum creatinine of  $\geq 26.5$   $\mu\text{mol/L}$  within 48 hours
- Increase in serum creatinine  $\geq 1.5$  times of baseline which is known or presumed to have occurred within the past 7 days
- Urine volume  $< 0.5$  ml/kg/h for 6 hours

Observe for eGFR trend in past 12 months, if stable trajectory with well patient, minimal albuminuria and unremarkable urine microscopy, this can be presumed to be CKD progression and given a routine review.

When encountering patients with significant eGFR drop ( $\geq 5$  - 10 depending on initial eGFR), evaluate for potential causes:

- Pre-renal: Dehydration, acute intercurrent illnesses, fluid losses
- Renal: New medications, traditional or over-the-counter medications, drug-drug interactions, flare of autoimmune conditions
- Post-renal: Obstructive uropathy (strictures, stones, enlarged prostate, blocked catheter), urinary tract infections

Assess and evaluate patients for symptoms and signs of fluid overload.

Repeat non-fasting sodium, potassium, creatinine within 3 - 7 days. Advise patient on low protein diet 12 hours prior to the test. Consider checking urine microscopy and albumin to optimise medical therapy. The threshold for referral is lower for patients with more advanced CKD (eGFR drop from 20 to 10 may warrant emergency visit due to significant creatinine rise).

### **Recommended ACEi / ARB to initiate for CKD patients**

A recent qualitative study conducted by MOH ACE has highlighted potential benefits of initiating Losartan and Telmisartan for patients with CKD. These medications may offer advantages in terms of dosing simplicity, as they typically require fewer dose adjustments to reach maximal effectiveness and do not necessitate renal dose adjustments in advanced CKD, potentially streamlining management for healthcare providers. However, it is important to note that the blood pressure-lowering effects can vary among different ACEi / ARB, and we can continue to recommend individualized treatment plans that may include additional blood pressure-lowering therapies, to achieve the target blood pressure of 130/80 mmHg for CKD patients.

### **When to repeat urine albumin/creatinine ratio after initiation or dose adjustment of ACEi / ARB or SGLT2i**

Studies have shown that these medications can reduce albuminuria and improve kidney function 3 – 6 months after initiation or dose adjustment; repeating urine albumin/creatinine ratio in 3 – 6 months is therefore recommended to assess need for further optimisation.

### **Indication for repeat ACEi / ARB panel after switching ACEi / ARB**

When transitioning between ACEi / ARB within the same equivalent dose range as indicated in the ACEi / ARB dose equivalence table, it is usually not necessary to repeat the ACEi / ARB panel if the patient remains clinically stable. The risk of electrolyte imbalances or complications related to renal artery stenosis is generally lower in such cases. However, individual patient factors should always be considered, and clinical judgment should be exercised in determining the need for follow-up testing, such as in the elderly, concomitant heart failure, poorly controlled diabetes, or patients on diuretics.

| <b>Delisted</b>                                   | <b>Alternatives</b>                                                                         |                                                         |                                                         |                                                   |                                |                                |
|---------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|--------------------------------|--------------------------------|
| <b>Candesartan (NS)</b>                           | <b>Losartan (S2)</b>                                                                        | <b>Enalapril (S1)</b>                                   | <b>Lisinopril (S1)</b>                                  | <b>Valsartan (S2)</b>                             | <b>Telmisartan (S2)</b>        | <b>Irbesartan (S2)</b>         |
| Indications:<br>• Hypertension<br>• Heart Failure | Indications:<br>• Hypertension<br>• Heart Failure                                           | Indications:<br>• Hypertension<br>• Heart Failure       | Indications:<br>• Hypertension<br>• Heart Failure       | Indications:<br>• Hypertension<br>• Heart Failure | Indications:<br>• Hypertension | Indications:<br>• Hypertension |
| 4mg om                                            | 25mg om                                                                                     | 5mg om                                                  | 5mg om                                                  | 40mg om                                           | 20mg om                        | 75mg om                        |
| 8mg om                                            | 50mg om                                                                                     | 5mg bd                                                  | 10mg om                                                 | 80mg om                                           | 40mg om                        | 150mg om                       |
| 12mg om                                           | 75mg om                                                                                     | 7.5mg bd                                                | 15mg om                                                 | 120mg om                                          | 60mg om                        | 225mg om                       |
| 16mg om                                           | 100mg om                                                                                    | 10mg bd                                                 | 20mg om                                                 | 160mg om                                          | 80mg om                        | 300mg om                       |
| 32mg om                                           | 100mg om                                                                                    | Nil available information, consider switch to other ARB | Nil available information, consider switch to other ARB | 160mg bd, or 320mg om                             | 80mg om                        | 300mg om                       |
| <b>Maximum Dose: 32mg om</b>                      | <b>Maximum Dose: 100mg om</b><br>Note: Losartan dose increases to 150mg by specialists only | <b>Maximum Dose: 20mg bd</b>                            | <b>Maximum Dose: 40mg om</b>                            | <b>Maximum Dose: 160mg bd, or 320mg om</b>        | <b>Maximum Dose: 80mg om</b>   | <b>Maximum Dose: 300mg om</b>  |

*Compiled by NUHS Pharmacy, Drug Information Service, March 2025*

### **Starting SGLT2i when ACEi/ARB is not fully optimized**

The MOH ACE guidelines for CKD provided an approach to management:

- Recommendation 2 states to “Optimise blood pressure control and albuminuria management with an ACEi / ARB, and titrate to maximum tolerated dose as needed”
- Recommendation 3 suggests to “Add an SGLT2i to ACEi / ARB therapy for patients with CKD and persistent albuminuria, regardless of DM status.”

While guidelines suggest using ACE inhibitors or ARBs before SGLT2 inhibitors, flexibility can be helpful for patients who have challenges for frequent tests for ACEi / ARB dose adjustments. There are no direct comparisons between optimized ACEi / ARB therapy and combined ACEi / ARB + SGLT2 inhibitor therapy for early kidney disease outcomes. For DM patients, starting both treatments together may be especially helpful due to added benefits for blood sugar control and possible weight loss. For non-DM CKD patients, the decision to start SGLT2 inhibitors before optimizing ACEi / ARB treatment should be based on individual patient factors, potential long-term benefits, and costs.

### **Handling cessation of ACEi / ARB or SGLT2i post-hospital discharge**

ACEi / ARBs play a crucial role in managing CKD, offering significant benefits in albuminuria control and slowing CKD progression. Our NUP study on HALT-CKD patients revealed that discontinuation or dose reduction of ACEi / ARBs was associated with an almost twofold increased risk of progression to advanced CKD. Similarly, SGLT2i have demonstrated additional benefits in reducing the risk of end-stage renal failure, cardiovascular events, and renal mortality (DAPA-CKD, EMPA-KIDNEY).

Given these findings, we propose for clinically stable patients with potentially good health outcomes, to consider a stepwise re-initiation and uptitration of ACEi / ARB and SGLT2i therapies that they were previously on. **This approach aims to optimize control while closely monitoring for any adverse effects.** In cases where these medications have been discontinued during hospitalization without clear documentation of the rationale, we recommend clarifying with the inpatient team, to discuss the potential for their safe reintroduction.

### **Use of ACEi / ARB and SGLT2i in Elderly**

Research on ACEi / ARB use in elderly patients suggests comparable effectiveness and safety to younger populations. However, elderly patients remain vulnerable to drug interactions due to age-related physiological changes and increased medication use. Several studies have reported elevated risks of hospitalization associated with ACEi / ARB interactions, including combinations with potassium-sparing diuretics, sulfamethoxazole/trimethoprim, and loop diuretics. While ACEi / ARB therapy appears generally safe in the elderly, careful monitoring and awareness of potential drug interactions and adverse effects (postural hypotension) are crucial to minimize risks and

optimize treatment outcomes. In patients who have been on long-term ACEi / ARB with no major issues, these medications can be continued for control of CKD and blood pressure, individualized to the patient's function and health.

SGLT2i have shown efficacy and safety in elderly patients with type 2 diabetes. Studies have shown significant reductions in HbA1c, BMI, and albuminuria benefits without compromising on renal function (SOLD study). They also provide consistent relative risk reductions in cardiovascular mortality and heart failure hospitalisation across age groups, with potentially greater absolute risk reductions in elderly due to higher baseline risk. While well tolerated, caution is advised in frail elderly, especially with risk of volume depletion. Hence, it is important to contextualize the potential long-term benefits individualized to the patient's current function and health.

### **Use of ACEi/ARB and SGLT2i in Advanced Chronic Kidney Disease**

The use of ACEi/ARB in advanced CKD remains controversial, with minimal data supporting the initiation of ACEi/ARB in advanced CKD patients. When ACEi/ARB were started in early CKD, continuing or discontinuing ACEi/ARB in advanced CKD was not associated with significant difference in rate of eGFR decline or adverse events (STOP ACEi trial). However, certain ACEi/ARB will need to be renally adjusted to ensure safety to patients (e.g. Captopril, Enalapril, Perindopril, etc). ACEi/ARB with renally adjusted doses can, therefore, be continued for patients in advanced CKD, individualized to the patient's current function, health, and outcomes of overall therapy, unless hyperkalaemia, hypotension, or unusually rapid worsening of eGFR occurs.

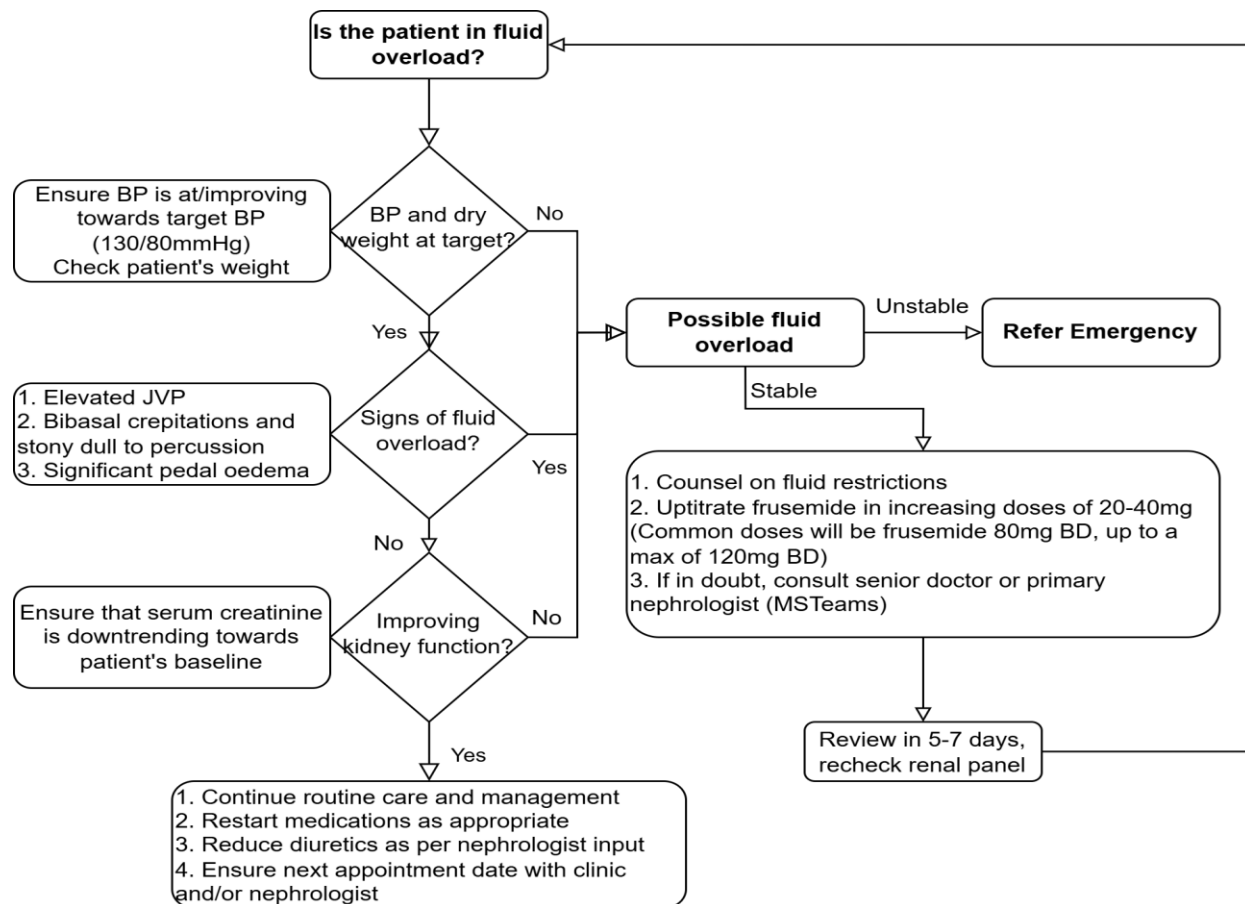
The use of SGLT2i in advanced CKD is well-established for Dapagliflozin (DAPA-CKD), where it can be continued until dialysis. We assume a class effect with Empagliflozin, although no major studies have been conducted regarding its use in end stage renal failure. Real-world studies have demonstrated improved kidney function, glycaemic control, and reduced adverse kidney-related and cardiovascular events in patients with advanced CKD. While initial safety concerns existed, accumulating clinical data are reassuring.

### **Fluid Overload Management of advanced CKD patients**

NUP physicians may be required to perform early reviews for advanced CKD patients post-discharge for fluid overload.

The following table serves as a guideline in managing these patients. Specifically, the tasks of the reviewing doctor include:

- Clinically assess fluid status, including BP and weight
- Review renal panel performed prior to consult
- Manage and adjust diuretics as appropriate
- Ensure appropriate TCU given to patient with safety net advice



## REFERRAL CRITERIA

| Clinical Problem                                                                                                                                                                                                                                                                         | Initial Management                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Disposition                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Rise in creatinine</b> <ul style="list-style-type: none"> <li>➤ More than 2x baseline</li> <li>➤ More than 1.5x baseline <ul style="list-style-type: none"> <li>○ No change in ACE-I/ARB in last 3 months</li> <li>○ Increase in ACE-I/ARB in last 3 months</li> </ul> </li> </ul> | Repeat within 3 – 7 days, if remains elevated----><br><br>Repeat within 3 – 7 days, if remains elevated----><br><br>Stop or decrease ACE-I/ARB<br>Recheck creatinine within 2 weeks <ul style="list-style-type: none"> <li>➤ Creatinine back to baseline? <ul style="list-style-type: none"> <li>○ YES -----&gt;</li> <li>○ NO &amp; &gt;30% rise -----&gt;</li> <li>○ Initial Management NO &amp; &gt;50% rise (1.5x to &lt; 2x baseline) -----&gt;</li> </ul> </li> </ul> | <b>Refer A&amp;E</b><br><br><b>Direct access Nephro</b><br><br><br>Routine Nephro appt<br><b>Early Nephro appt</b><br><br><b>Direct access Nephro</b> |
| <b>2. Hyperkalaemia</b> <ul style="list-style-type: none"> <li>➤ <math>K^+ \geq 6</math></li> <li>➤ <math>K^+ 5.6-5.9</math></li> </ul>                                                                                                                                                  | -----><br>Hyperkalaemia management as per protocol<br>Repeat s. $K^+$ within 1 week<br>If s. $K^+$ remains 5.6 – 5.9 ----->                                                                                                                                                                                                                                                                                                                                                 | <b>Refer A&amp;E</b><br><br><b>Direct access Nephro</b>                                                                                               |
| <b>3. “Fluid overload” in CKD G3 to G5</b> <ul style="list-style-type: none"> <li>➤ “Fluid overload” diagnosis in CKD G5 despite dose of loop diuretic equivalent to a minimum of 120mg daily</li> <li>➤ “Fluid overload” diagnosis in CKD G3 - G4 despite loop diuretic</li> </ul>      | -----><br><br>----->                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Refer A&amp;E</b><br><br><b>Early Nephro appt</b>                                                                                                  |
| <b>4. CKD</b> <ul style="list-style-type: none"> <li>➤ CKD G5 on 2 occasions over 90-day period and asymptomatic</li> <li>➤ CKD G3B – G4, eGFR decline &gt; 5mL/min/1.73m<sup>2</sup> <b>over 3 months</b></li> </ul>                                                                    | -----><br><br>----->                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Early Nephro appt</b><br><br><b>Early Nephro appt</b>                                                                                              |

|                                                                                                                                            |                                                                                                  |                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| ➤ <b>Average eGFR decline &gt; 10 mL/min/1.73m<sup>2</sup> over 12 months</b>                                                              | ----->                                                                                           | <b>Early Nephro appt</b>                                                             |
| <b>5. Proteinuria</b>                                                                                                                      |                                                                                                  |                                                                                      |
| ➤ Elevated creatinine/ low eGFR                                                                                                            | ----->                                                                                           | Refer to point 1 or 2. For A+E/Direct access/Early Nephro appt. Otherwise see below. |
| ➤ New onset nephrotic range proteinuria<br>UPCR > 300 mg/mmol or UACR > 200 mg/mmol on 2 samples                                           |                                                                                                  |                                                                                      |
| ○ Non-Diabetic                                                                                                                             | ----->                                                                                           | <b>Direct access Nephro</b>                                                          |
| ○ Diabetic                                                                                                                                 | ----->                                                                                           | Routine Nephro appt                                                                  |
| ➤ UPCR > 100 or UACR > 70mg/mmol on 2 samples                                                                                              |                                                                                                  |                                                                                      |
| ○ Haematuria present (2 samples with ≥ 4 RBCs/μl or ≥ 3 RBCs/hpf)                                                                          | ----->                                                                                           | <b>Early Nephro appt</b>                                                             |
| ○ No haematuria                                                                                                                            |                                                                                                  |                                                                                      |
| ▪ Non-diabetic                                                                                                                             | ----->                                                                                           | Routine Nephro appt                                                                  |
| ▪ Diabetic                                                                                                                                 |                                                                                                  |                                                                                      |
|                                                                                                                                            | Optimise ACE-I/ARB                                                                               |                                                                                      |
|                                                                                                                                            | ➤ Persistent uPCR > 100 or uACR > 70 mg/mmol on 2 samples ----->                                 | Routine Nephro appt                                                                  |
| ➤ UPCR > 50 or UACR > 30 mg/mmol on 2 samples + <b>haematuria</b> (2 samples with ≥ 4 RBC/μl or ≥ 3 RBC/hpf), no hypertension / DM         | ----->                                                                                           | Routine Nephro appt                                                                  |
| <b>6. RPGN / Nephritic syndrome</b>                                                                                                        |                                                                                                  |                                                                                      |
| ➤ <b>Rapidly progressive glomerulonephritis (RPGN) suspected</b> (haematuria, oligouria, hypertension and rapid rise in Cr at least > 20%) | ----->                                                                                           | <b>Refer A&amp;E</b>                                                                 |
|                                                                                                                                            | Advise patients to monitor BP, urine output and haematuria in event of progression to RPGN ----> | <b>Direct access Nephro</b>                                                          |

|                                                                                                                                                                                                                                                                                                                                                                                             |                                                    |                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------|
| ➤ <b>Nephritic syndrome with stable serum creatinine &lt; 20% difference over 90 days</b>                                                                                                                                                                                                                                                                                                   |                                                    |                                                                                        |
| <b>7. CKD with uncontrolled hypertension</b> <ul style="list-style-type: none"> <li>➤ Systolic BP &gt; 200 mmHg or Diastolic BP &gt; 120 mmHg</li> <li>➤ Systolic BP &gt; 180 mm Hg, diastolic BP &gt; 100 mmHg despite 4 anti-hypertensive agents at moderately high dosages, including a diuretic</li> <li>➤ BP &gt; 150/90 mmHg on maximal doses of 3 antihypertensive agents</li> </ul> | <p>-----&gt;</p> <p>-----&gt;</p> <p>-----&gt;</p> | <p><b>Refer A&amp;E</b></p> <p><b>Early Nephro appt</b></p> <p>Routine Nephro appt</p> |
| <b>8. CKD with poorly controlled risk factors despite maximum doses of all standard medications</b><br><b>Risk factors include: BP, DM, proteinuria</b>                                                                                                                                                                                                                                     | <p>-----&gt;</p>                                   | <p>Routine Nephro appt</p>                                                             |
| <b>9. Suspected of having rare / genetic causes of CKD</b><br>E.g. Polycystic kidney disease                                                                                                                                                                                                                                                                                                | <p>-----&gt;</p>                                   | <p>Routine Nephro appt</p>                                                             |
| <b>10. Anaemia due to CKD</b>                                                                                                                                                                                                                                                                                                                                                               | <p>-----&gt;</p>                                   | <p>Refer to <a href="#">NUP Anaemia CPG</a> or <a href="#">NUP CKD Shared Care</a></p> |

## RECOMMENDED CARE COMPONENTS FOR CHRONIC KIDNEY DISEASE

| Recommended Care Components                                                               | Minimum Frequency*                                                                                                       | Remarks                                                                                                                  |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Blood Pressure measurement~                                                               | Twice a year                                                                                                             | ACE-I and ARBs should be used for BP control when proteinuria is present                                                 |
| Weight and BMI                                                                            | Twice a year                                                                                                             |                                                                                                                          |
| Lipid profile                                                                             | Annually                                                                                                                 |                                                                                                                          |
| Diabetes screening                                                                        | Annually                                                                                                                 | Or more frequent in pre-diabetes or diabetes                                                                             |
| Kidney Function – Sodium, Potassium, Creatinine and eGFR~                                 | Twice a year                                                                                                             |                                                                                                                          |
| Albuminuria – Urine Protein: Creatinine Ratio (uPCR) or Albumin: Creatinine Ratio (uACR)~ | Twice a year                                                                                                             |                                                                                                                          |
| Smoking assessment                                                                        | Annually for smokers                                                                                                     | Once-off for non-smokers unless change in smoking habit                                                                  |
| Influenza Vaccination <sup>#</sup>                                                        | Annually or per season                                                                                                   | As recommended under the National Adult Immunisation Schedule (NAIS) and National Childhood Immunisation Schedule (NCIS) |
| Pneumococcal, Herpes Zoster, COVID-19 Vaccinations                                        | As recommended under the National Adult Immunisation Schedule (NAIS) and National Childhood Immunisation Schedule (NCIS) |                                                                                                                          |
| Hepatitis B Vaccinations                                                                  | As directed by Nephrology                                                                                                |                                                                                                                          |

\* More frequently if clinically indicated

~ These are reportable clinical indicators

- FBC, adjusted Calcium, Phosphate and Bicarbonate should be routinely monitored in patients with advanced CKD

## **ABBREVIATIONS**

|        |                                           |
|--------|-------------------------------------------|
| ACE-I  | Angiotensin-Converting Enzyme Inhibitors  |
| ARB    | Angiotensin II Receptor Blockers          |
| CKD    | Chronic kidney disease                    |
| CV     | Cardiovascular                            |
| eGFR   | Estimated glomerular filtration rate      |
| ESRF   | End-stage renal failure                   |
| GN     | Glomerulonephritis                        |
| NAIS   | National Adult Immunisation Schedule      |
| NCIS   | National Childhood Immunisation Schedule  |
| SGLT2i | Sodium/glucose cotransporter-2 inhibitors |
| uACR   | Urine Albumin: Creatinine Ratio           |
| uPCR   | Urine Protein: Creatinine Ratio           |

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## **CKD SHARED CARE PROGRAMME**

CKD Shared Care Programme is a national programme that aims to optimise management of patients with **CKD 3B** by anchoring part of their care in the community.

CKD 3B patients may be referred from Renal SOCs to NUP for shared care.

- Referral will be via the usual “Referral to Polyclinics” Epic template for SOC’s on Epic
- Referral from non-Epic SOC’s (SingHealth → via hardcopy memo)
- Relevant follow-up instructions for the shared care will be provided by the referring SOC

### **Instructions for Admin Doctors handling CKD Shared Care referrals**

CKD Shared Care patients are referred to NUP via the Stepdown care workflow, via “Referral to Polyclinics” Epic order to respective clinics.

- Review the referral carefully, order blood tests and TCU as per instructions in the referral (typically in **6** months)
- If referred patient is **NOT** CKD 3B, to continue co-management, reject Shared Care referral (Contact primary nephrologist via Teams), document on Epic
- If referred patient is **NOT** teamlet or HSG-enrolled with NUP, reject referral (Contact primary nephrologist via Teams), document on Epic

## **Instructions for Reviewing Doctors Seeing CKD Shared Care Patients**

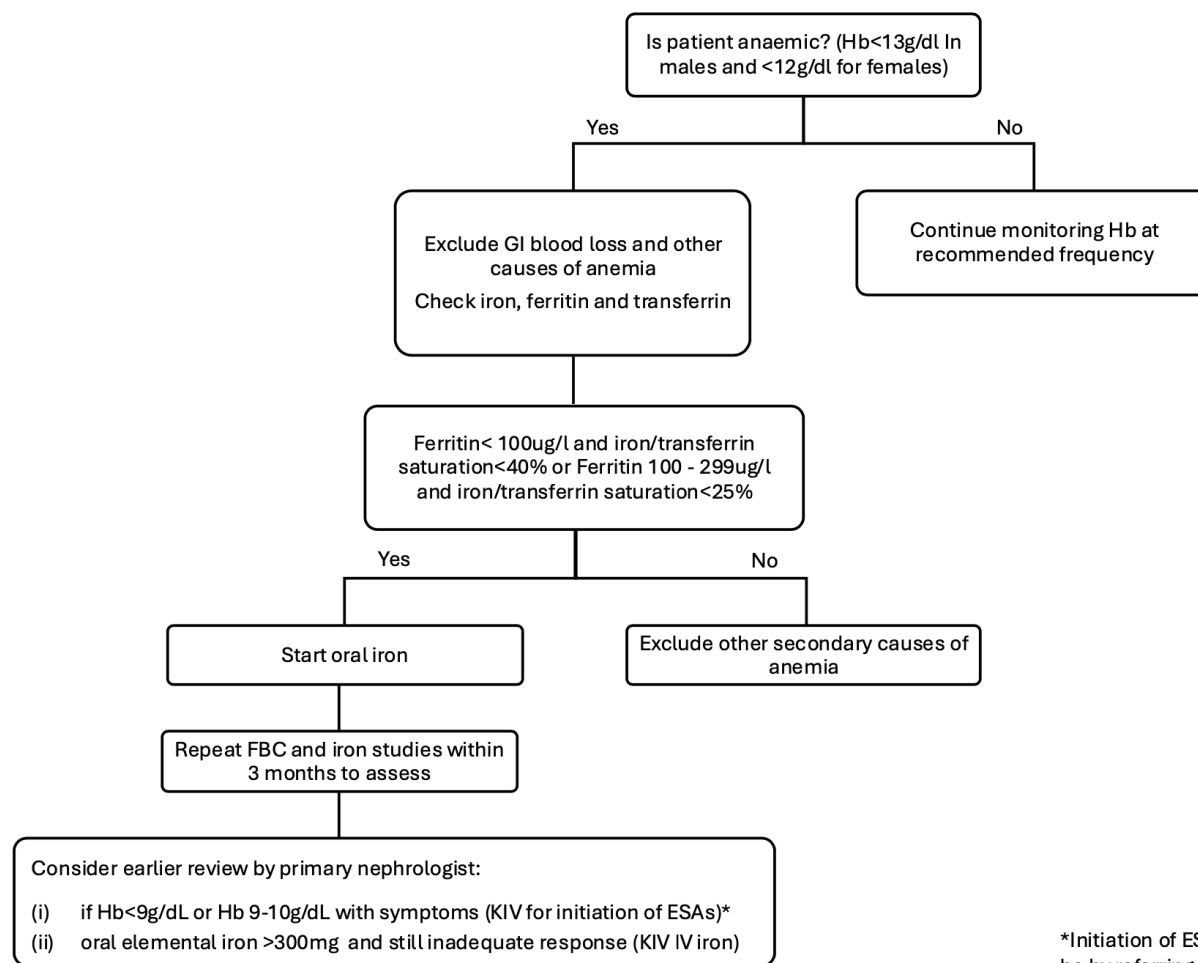
- Review the referral from Renal SOC carefully and review blood tests performed prior to consultation
- Use suggested checklist (.NUPDRCKDSHAREDCARE) to manage advanced CKD
- Book subsequent NUP consults, ensure that patient has a Nephrology TCU
- Document that patient is under CKD Shared Care in Dr Consult and/or Care Coordination Notes

### **Escalation criteria for CKD Shared Care – to contact primary nephrologist via Teams**

- Acute kidney injury
- Hb < 9g/dl despite iron supplementation, or Hb 9-10 g/dl with symptoms and non-renal causes have been excluded (for consideration of initiation of ESA)
- Rapid decline of eGFR of > 5ml/min/1.73m<sup>2</sup>/year
- Bicarbonate level < 18mmol/L despite oral sodium bicarbonate
- Phosphate level persistently elevated at > 1.8mmol/L despite dietary phosphate restriction and initiation of phosphate binders
- Suspicion of new onset glomerulonephritis or nephrotic syndrome
- Persistent hyperkalemia K > 5.5 mmol despite medical treatment
- Worsening proteinuria > 1g/day and UACR > 700mg/g despite optimized dose of ACEi / ARB and SGLT2i

### **Management of Anaemia in CKD Shared Care**

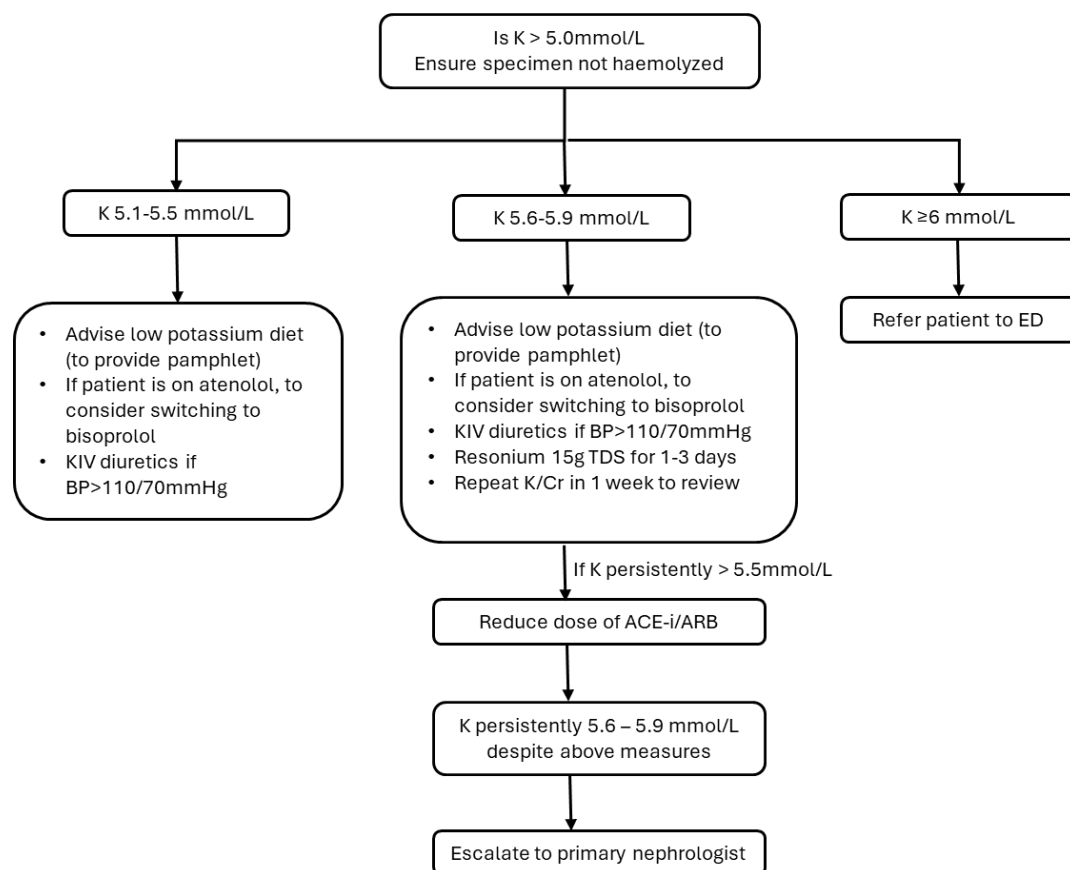
- If Hb is low, evaluate for secondary causes of anaemia
- Perform iron panel, start iron replacement and up-titrate as tolerated
- Iron replacement: Either with iron polymaltose (100mg elemental iron per tab) or Sangobion (ferrous gluconate) (30mg elemental iron per capsule)
- If Hb is persistently < 9g/dL despite maximum tolerated iron, refer Renal for consideration of ESA / EPO



\*Initiation of ESAs and dose titration of ESAs would be by referring nephrologist

## Management of Hyperkalaemia in CKD Shared Care

**FOR CKD 3B under CKD  
Shared Care Programme only**



If persistent hyperkalaemia  $\geq 5.6$  mmol/L, consider:

- Reducing dose of ACEi / ARB
- Switching atenolol to bisoprolol
- Check and correct acidosis: If serum bicarbonate 18-22 mmol/L, start oral sodium bicarbonate (see acidosis workflow)
- Escalate to nephrologist for consideration of chronic hyperkalaemia treatment (e.g. Sodium zirconium cyclosilicate [Lokelma])

## **Management of Metabolic Acidosis in CKD Shared Care**

For CKD Shared Care, sodium bicarbonate is initiated by nephrologists. Do **NOT** perform serum bicarbonate except in management of stable CKD Shared Care patients every 6 monthly.

The prevalence of metabolic acidosis was 13% in stage 3 and 37% in stage 4 CKD. In CKD management, we aim serum bicarbonate  $\geq 18$  mmol/l. Sodium bicarbonate is used in the management of metabolic acidosis in advanced CKD. Avoid sodium bicarbonate if patient is fluid overloaded or hypernatraemic ( $\text{Na} > 145\text{mmol/L}$ ).

Doctors should evaluate for other possible causes of metabolic acidosis:

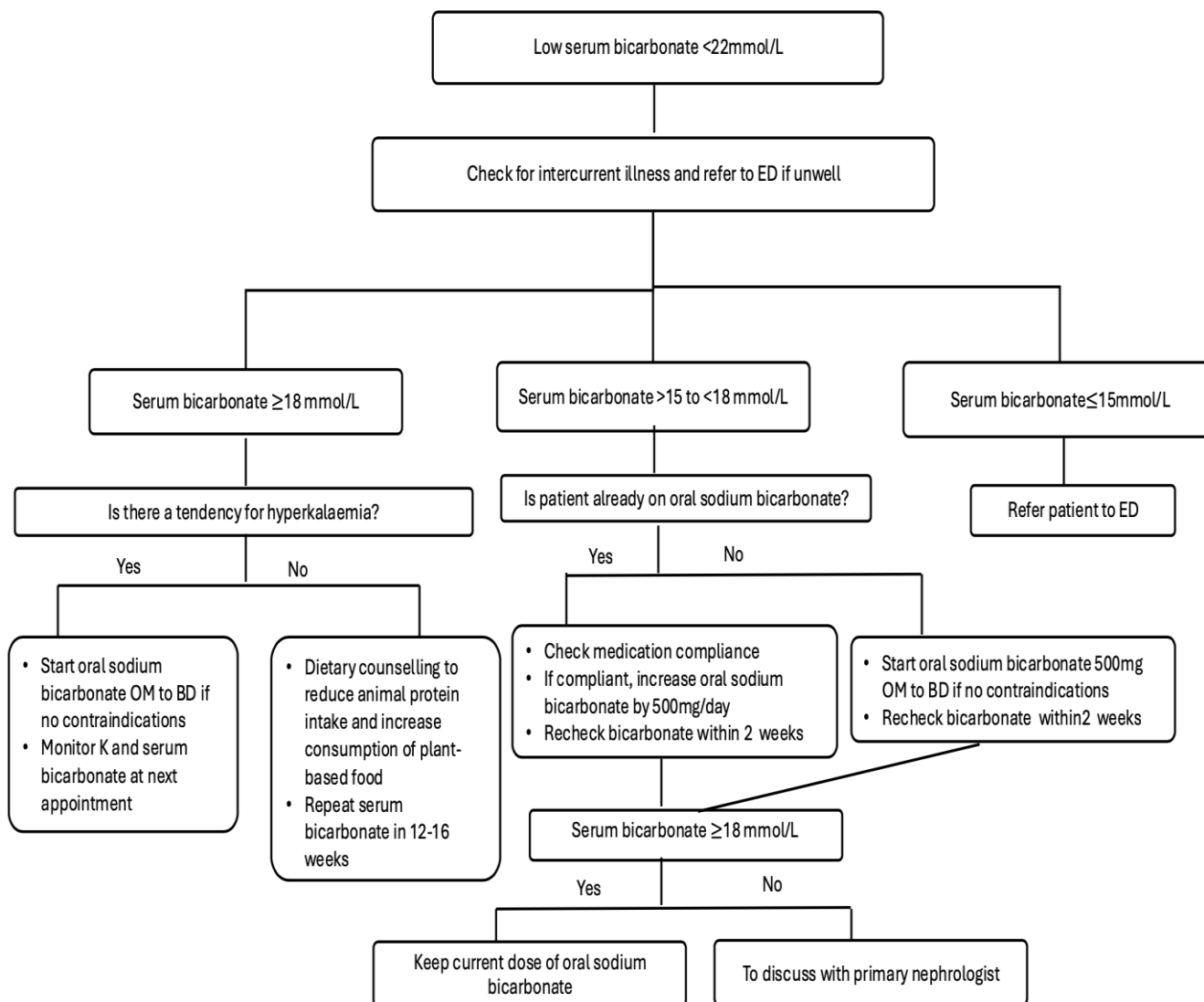
- Diabetes: Diabetic ketoacidosis, hyperosmolar hyperglycaemic state
- Nephrology: Advanced CKD, AKI, renal tubular acidosis, nephrotic syndrome
- Gastrointestinal: Intestinal obstruction, severe diarrhoea
- Starvation ketoacidosis
- Toxins: Alcohol, drugs, aspirin overdose
- Sepsis, shock, severe burns

If low bicarbonate was due to advanced CKD:

- Start sodium bicarbonate 500mg BD
- Monitor BP and for pedal oedema
- Recheck bicarbonate in 1-2 weeks

If patient is already on sodium bicarbonate:

- Increase dose by 500mg (e.g. 500mg BD  $\rightarrow$  1g OM, 500mg ON, up to maximum of 3g/day)
- Recheck in 1-2 weeks



Tendency for hyperkalaemia refers to frequent episodes of patients with  $K > 5.1 \text{ mmol/L}$  or with a single episode of hyperkalaemia episode ( $K > 5.5 \text{ mmol/L}$ ) requiring correction with Resonium.

### Management of CKD Mineral Bone Disease (MBD) in CKD Shared Care

- Initial management should be that of dietary phosphate restriction. Limit intake of high phosphate foods (fish with edible bones (sardines), anchovies (*ikan bilis*), dried shrimps, organ meats, meat / fish bones, processed foods, malt-based drinks, cola-based soft drinks, chocolate and fish / prawn crackers) and choose low phosphate alternatives.
- Phosphate binders can be prescribed if phosphate is persistently elevated despite dietary phosphate restriction.
- If calcium is elevated with low phosphate, to consider referral to nephrologist for non-calcium-based phosphate binders

|                           | Calcium acetate                                                                                                                                                                                                                           | Calcium carbonate                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                           |                                                                                                                                                         |  |
| Elemental calcium content | 169mg/667mg tab                                                                                                                                                                                                                           | 250mg/625mg tablet<br>500mg/1.25g tablet                                            |
| With respect to food      | Swallow tablet with meals                                                                                                                                                                                                                 | Chew/crush with meals                                                               |
| Drug interaction          | Thyroxine: space 2 hours before binders<br>Ciprofloxacin: space 2 hours before or 6hours after binders<br>Levofloxacin: space 2 hours before or after binders<br>Tetracyclines (Doxycycline, Minocycline): space 2 hours or after binders |                                                                                     |

